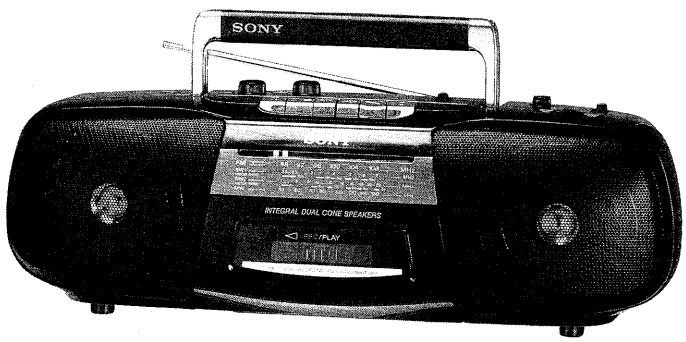
CFS-B5S

SERVICE MANUAL

E Model
East European Model



Model Name Using Similar Mechanism	CFM-155S
Tape Transport Mechanism Type	MF-B5S

SPECIFICATIONS

Frequency range
- Saudi Arabia model
  - FM: 87.6–107 MHz
  - MW: 530–1,605 kHz
  - SW1: 2.3–7 MHz
  - SW2: 7–22 MHz
- East European model
  - FM: 65–108 MHz
  - MW: 531–1,602 kHz
  - SW1: 2.3–7 MHz
  - SW2: 7–22 MHz
- Other models
  - FM: 87.6–108 MHz
  - MW: 530–1,605 kHz
  - SW1: 2.3–7 MHz
  - SW2: 7–22 MHz

Aerials
- FM/SW: Telescopic aerial
- MW: Built-in ferrite bar aerial

Recording system
- 4-track, 2-channel stereo

Frequency response
- 80–10,000 Hz

Speakers
- Full range: 10 cm (4 inches) dia., 3.2 ohms cone type x 2

Maximum power output
- 2.5 W + 2.5 W

Output
- Headphones jack (stereo minijack) for 16–68 ohms inpedance headphones

Power requirements
- 100–120 V, 220–240 V AC, 50/60Hz selectable
- 9 V DC, six R20 (size D) batteries

Power consumption
- 14 W AC operation

Battery life

Batteries	FM recording	Tape playback
Sony SUM-1 (NS)	Approx. 16 hours	Approx. 8 hours
Sony AM1 alkaline	Approx. 36 hours	Approx. 21 hours

Dimensions
- Approx. 470 x 142.5 x 159 mm (w/h/d) (18 5/8 x 5 5/8 x 6 3/8 inches) including projecting parts

Mass
- Approx. 3.0 kg (6 lb 10 oz) including batteries

Accessory supplied
- Mains lead (1)

Design and specifications are subject to change without notice.

RADIO CASSETTE-CORDER

SONY®

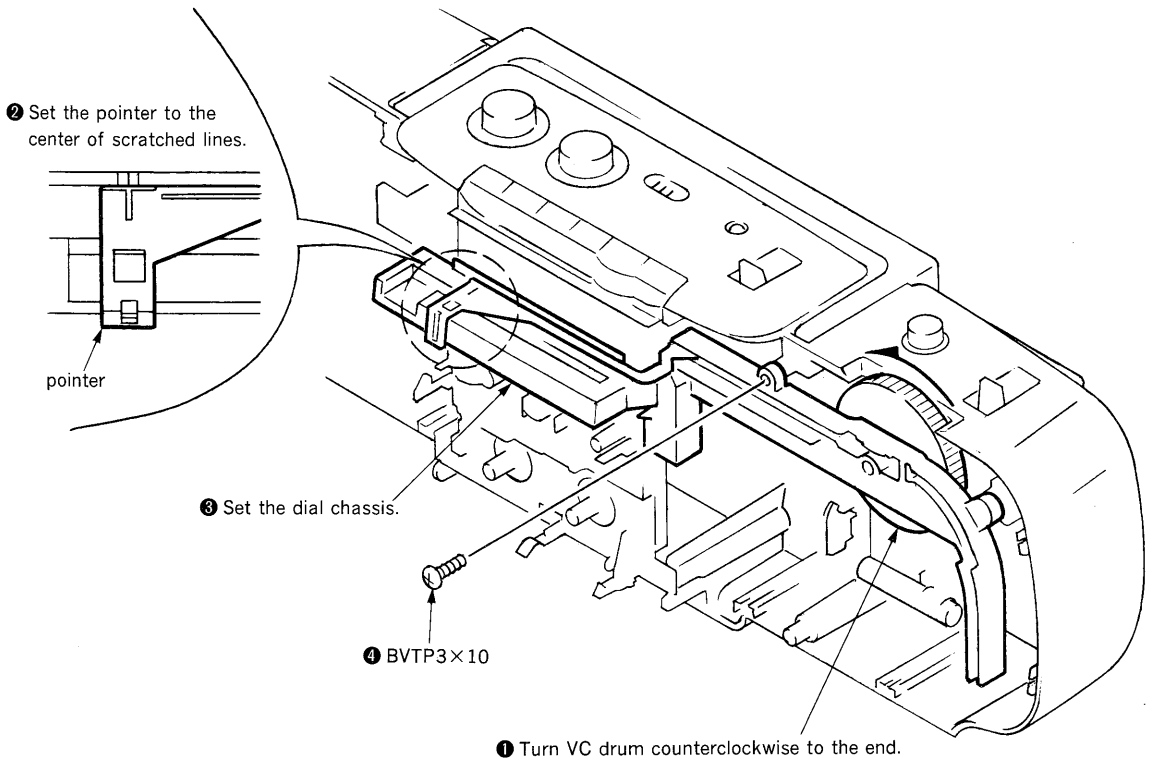
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SAFETY-RELATED COMPONENT WARNING!!

COMPONENTS IDENTIFIED BY MARK ⚠ OR DOTTED LINE WITH MARK ⚠ ON THE SCHEMATIC DIAGRAMS AND IN THE PARTS LIST ARE CRITICAL TO SAFE OPERATION. REPLACE THESE COMPONENTS WITH SONY PARTS WHOSE PART NUMBERS APPEAR AS SHOWN IN THIS MANUAL OR IN SUPPLEMENTS PUBLISHED BY SONY.

SECTION 1
DIAL POINTER SETTING



SECTION 2
MECHANICAL ADJUSTMENTS

PRECAUTION

1. Clean the following parts with a denatured-alcohol-moistened swab :
   record/playback head — pinch roller
   erase head — rubber belts
   capstan — idlers
2. Demagnetize the record/playback head with a head demagnetizer. (Do not bring the head demagnetizer close to the erase head.)
3. Do not use a magnetized screwdriver for the adjustments.
4. After the adjustments, apply suitable locking compound to the parts adjusted.
5. The adjustments should be performed with the rated power supply voltage unless otherwise noted.

Torque Measurement

Torque	Torque meter	Meter reading
Forward	CQ-102C	25−60g•cm (0.35−0.83 oz•inch)
Forward back tension	CQ-102C	1−5g•cm (0.014−0.069 oz•inch)
Fast Forward	CQ-201B	55−140g•cm (0.76−1.94 oz•inch)
Rewind	CQ-201B	55−140g•cm (0.76−1.94 oz•inch)

Tape Tension Measurement

Mode	Meter	Meter Reading
Forward	CQ-403A	more than 170g (more than 6.00 oz)

SECTION 3
ELECTRICAL ADJUSTMENTS

3-1. TAPE RECORDER SECTION 0dB=0.775V

Standard Output Level

Output terminal	HP OUT
load impedance	32Ω
output signal level	0.25V (−10dB)

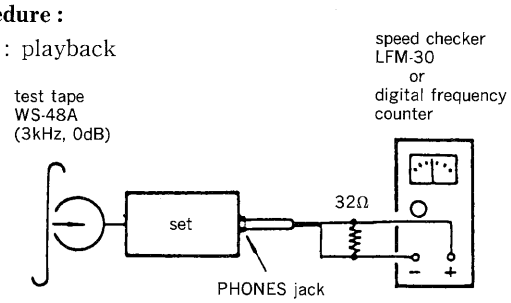
Test Tape

Type	Signal	Used for
WS-48A	3kHz, 0dB	tape speed adjustment
P-4-A063	6.3kHz, −10dB	head azimuth adjustment

Tape Speed Adjustment

Procedure :

Mode : playback

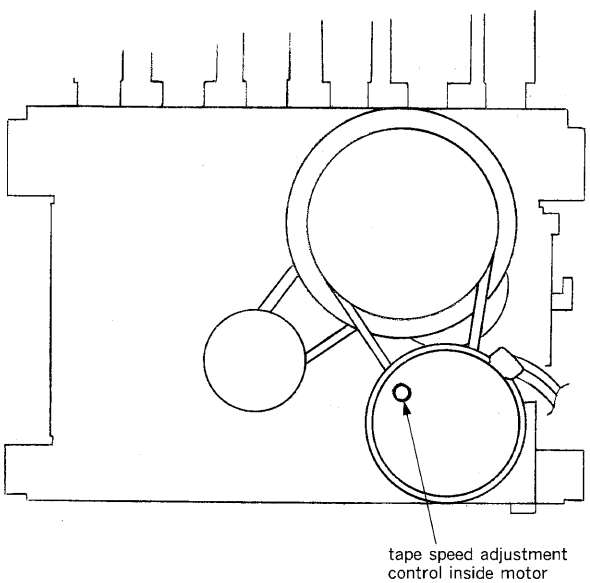


Adjustment Value :

Speed checker	Digital frequency counter
−1% to +1%	2,970 to 3,030Hz

Frequency difference between the beginning and the end of the tape should be within 1.5% (45Hz).

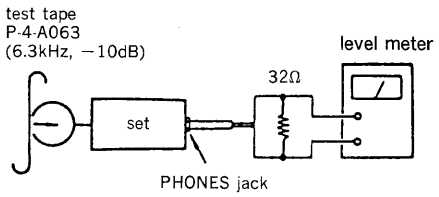
Adjustment Location :



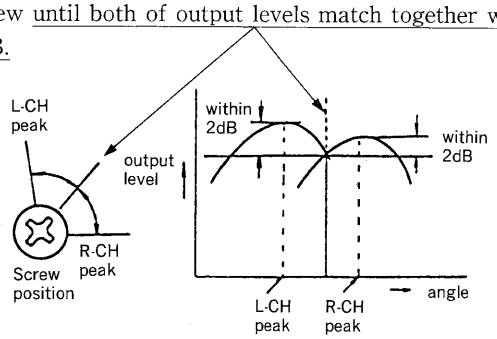
Record/Playback Head Azimuth Adjustment

Procedure :

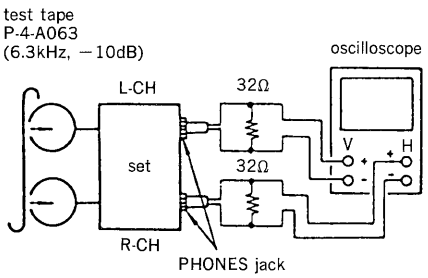
1. Mode : playback

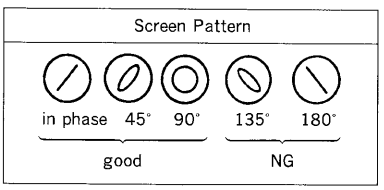


2. Turn the adjustment screw for the maximum output levels. If these levels do not match, turn the adjustment screw until both of output levels match together within 2dB.

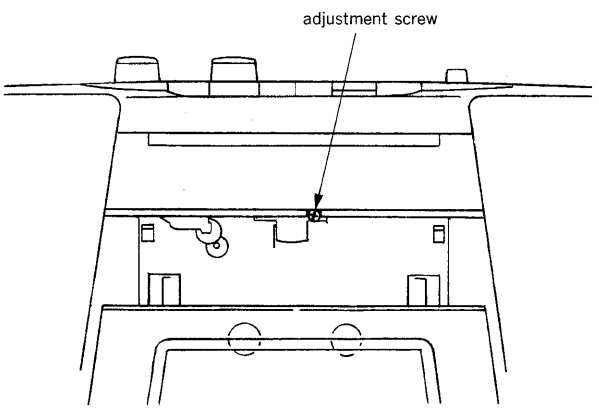


3. Phase Check
   Mode : playback





Adjustment Location :

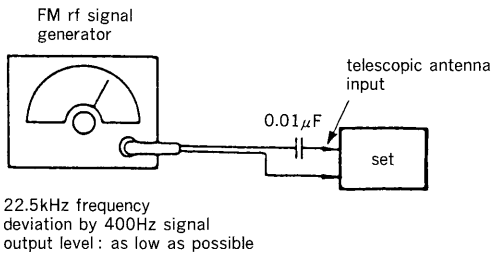


3-2. TUNER SECTION 0dB=1μV

• FM Section

Setting :

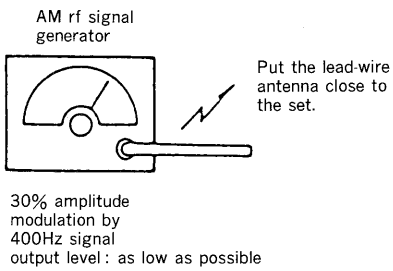
BAND switch : FM



• MW Section

Setting :

BAND switch : MW

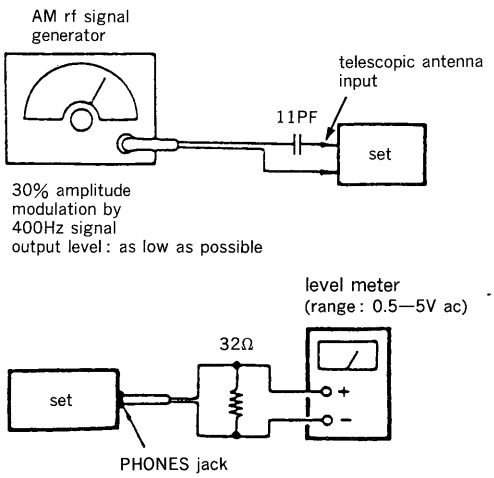


• SW Section

Setting :

BAND switch : SW1 or SW2

FINE TUNE control : mechanical mid



- Repeat the procedures in each adjustment several times, and the frequency coverage and tracking adjustments should be finally done by the trimmer capacitors.
- Remove telescopic antenna in FM section adjustments.

FM FREQUENCY COVERAGE ADJUSTMENT	
Adjust for a maximum reading on VTVM.	
L2	CT2
86.5MHz (87.35MHz) <65.0MHz>	109.5MHz (107.8MHz) <108.0MHz>

FM TRACKING ADJUSTMENT	
Adjust for a maximum reading on VTVM.	
L1	CT1
86.5MHz (87.35MHz) <65.0MHz>	109.5MHz (107.8MHz) <108.0MHz>

MW TRACKING ADJUSTMENT	
Adjust for a maximum reading on VTVM.	
L3	CT3
600kHz	1,400kHz

MW FREQUENCY COVERAGE ADJUSTMENT	
Adjust for a maximum reading on VTVM.	
L4	CT4
515kHz	1,680kHz

AM IF ADJUSTMENT
Adjust for a maximum reading on VTVM.
T1
455kHz

SW1 FREQUENCY COVERAGE ADJUSTMENT	
Adjust for a maximum reading on VTVM.	
L6	CT6
2.2MHz	7.3MHz

SW1 TRACKING ADJUSTMENT
Adjust for a maximum reading on VTVM.
L5
2.2MHz

SW2 FREQUENCY COVERAGE ADJUSTMENT	
Adjust for a maximum reading on VTVM.	
L8	CT10
6.8MHz	22.5MHz

SW2 TRACKING ADJUSTMENT
Adjust for a maximum reading on VTVM.
L7
6.8MHz

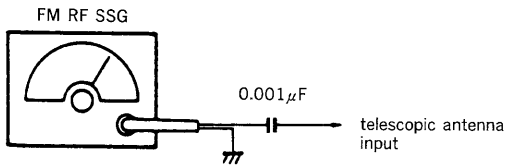
( ) : Saudi Arabia model

< > : East European model

Adjustment Location : See page 6.

FM VCO Adjustment

Setting :



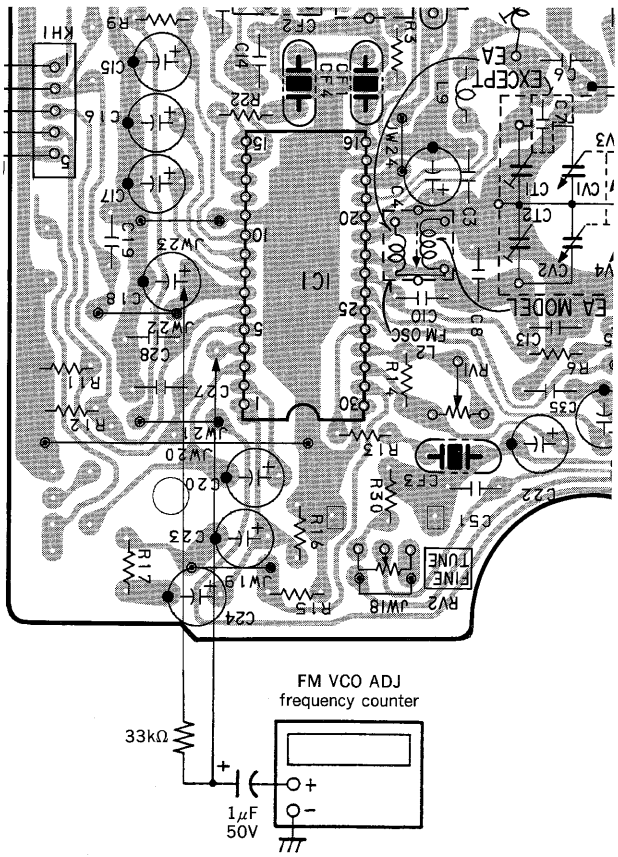
Modulation : no modulation
Output level : 0.1V (100dB)

Procedure :

1. Adjust RV1 so that the reading on the frequency counter becomes in 76.0kHz ±0.5kHz.

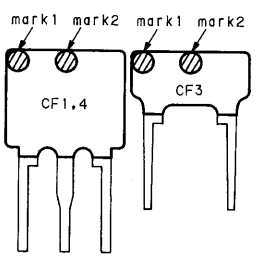
Adjustment Location : tuner board

TUNER BOARD (Conductor Side)



How To Changed The Ceramic Filters

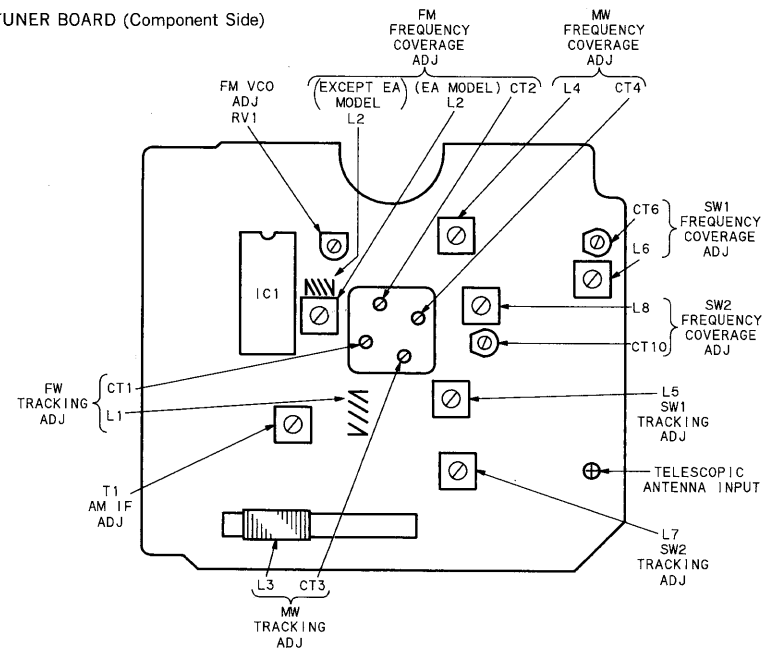
This model is used three ceramic filters of CF1, 3 and 4.
You must use same type of color marked ceramic filters in order to meet same specifications.
Therefore, the ceramic filter must change three pieces together since it's supply three pieces in one package as a spare parts.



mark1	mark2	Center Frequency
red	—	10.70MHz
blue	—	10.67MHz
orange	—	10.73MHz
black	—	10.64MHz
white	—	10.76MHz
white	white	10.75MHz
yellow	—	10.79MHz

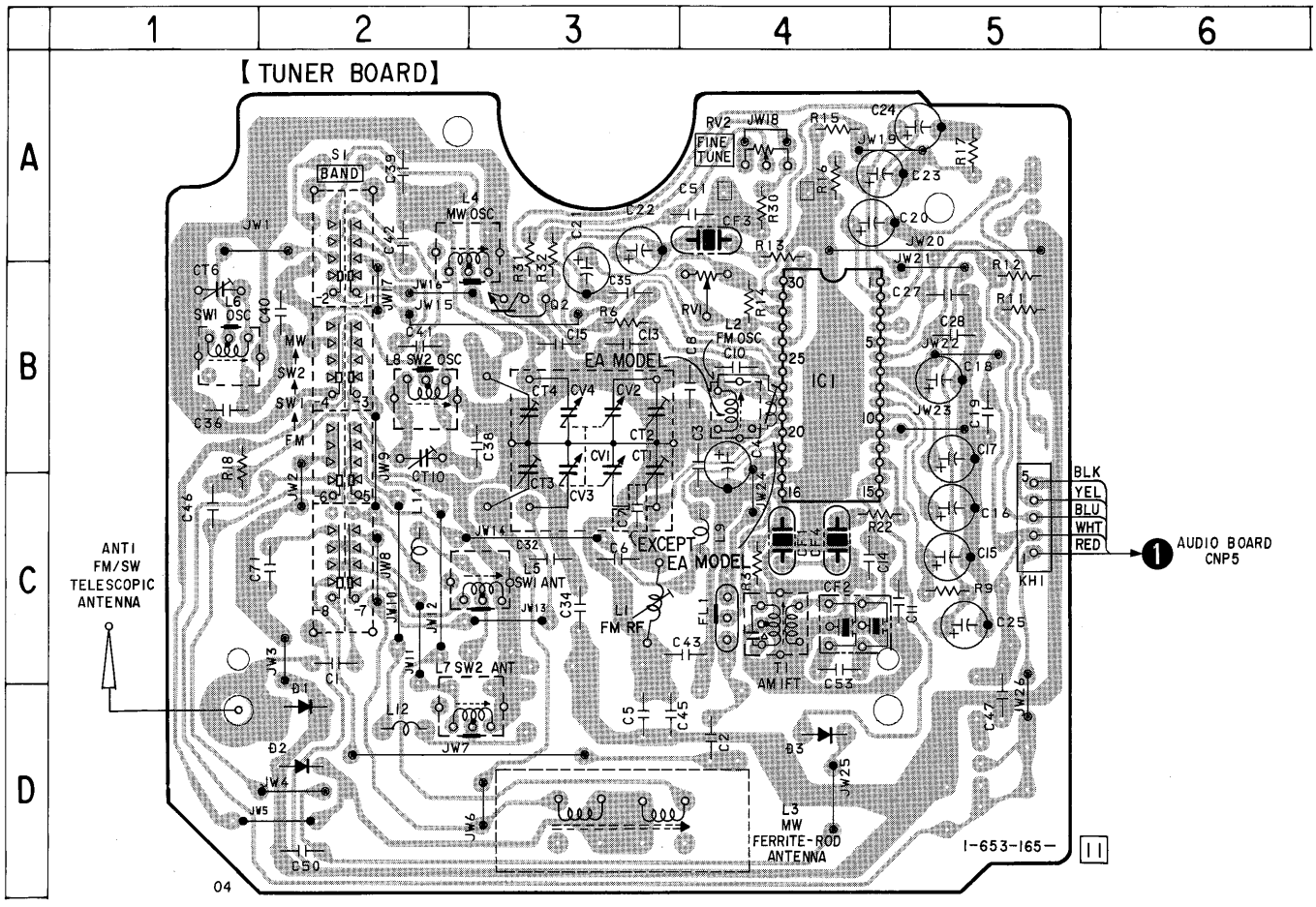
Adjustment Location :

TUNER BOARD (Component Side)



SECTION 4
DIAGRAMS

4-1. PRINTED WIRING BOARD—TUNER SECTION—



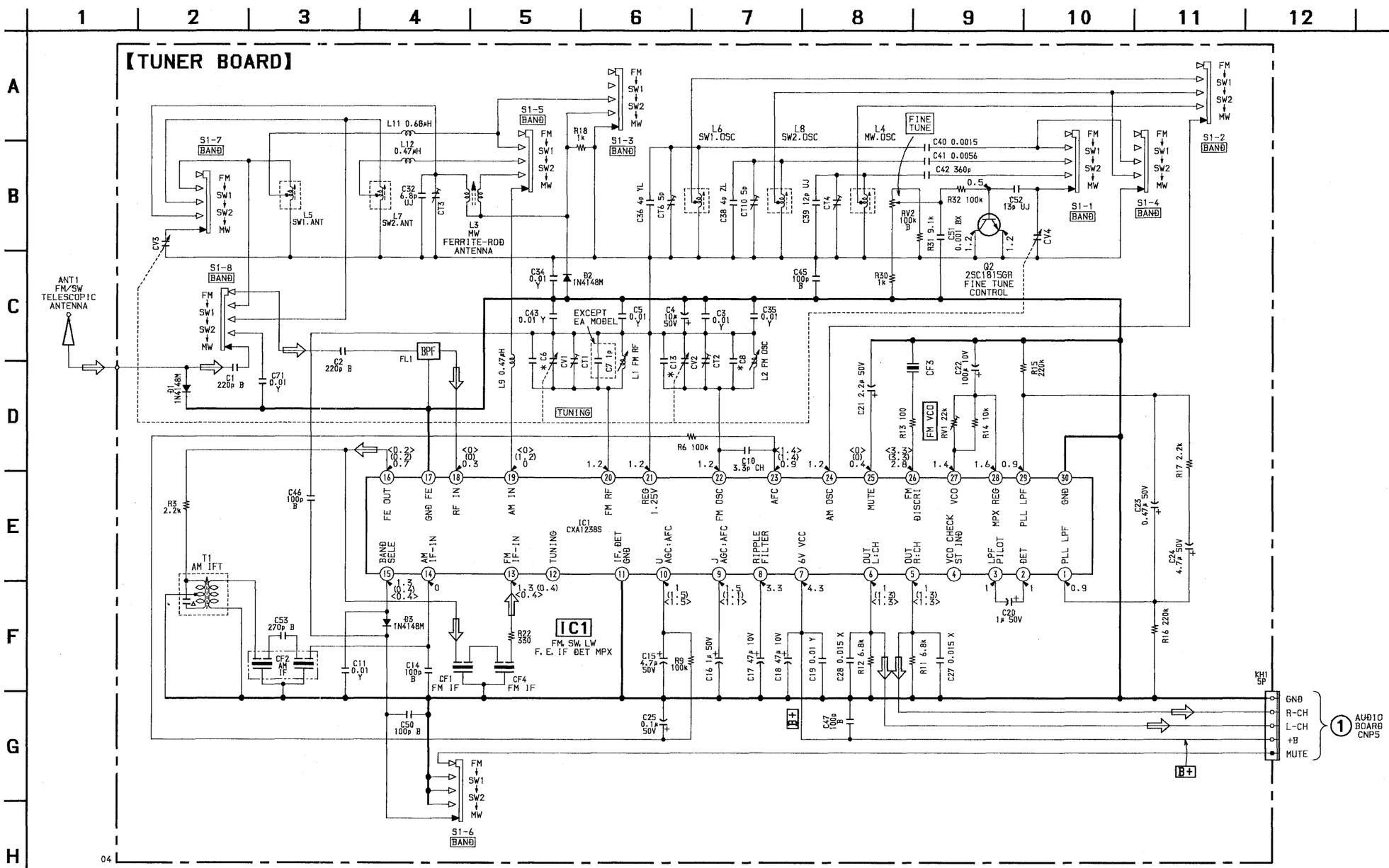
● Semiconductor Location

Ref. No.	Location
D1	D-2
D2	D-2
D3	D-4
IC1	B-4
Q2	B-3

Note :

- ○— : parts extracted from the component side.
- ⬚← indicates side identified with part number.
- ▩ : Pattern on the side which is seen.

4-2. SCHEMATIC DIAGRAM—TUNER SECTION— • Refer to page 16 for IC Block Diagram.



*	EXCEPT EA, EE MODEL	EA MODEL	EE MODEL
C6	18p SL	27p SL	8.2p SL
C8	18p SH	22p TH	10p UJ
C13	5.6p CH	6.8p CH	5.6p CH

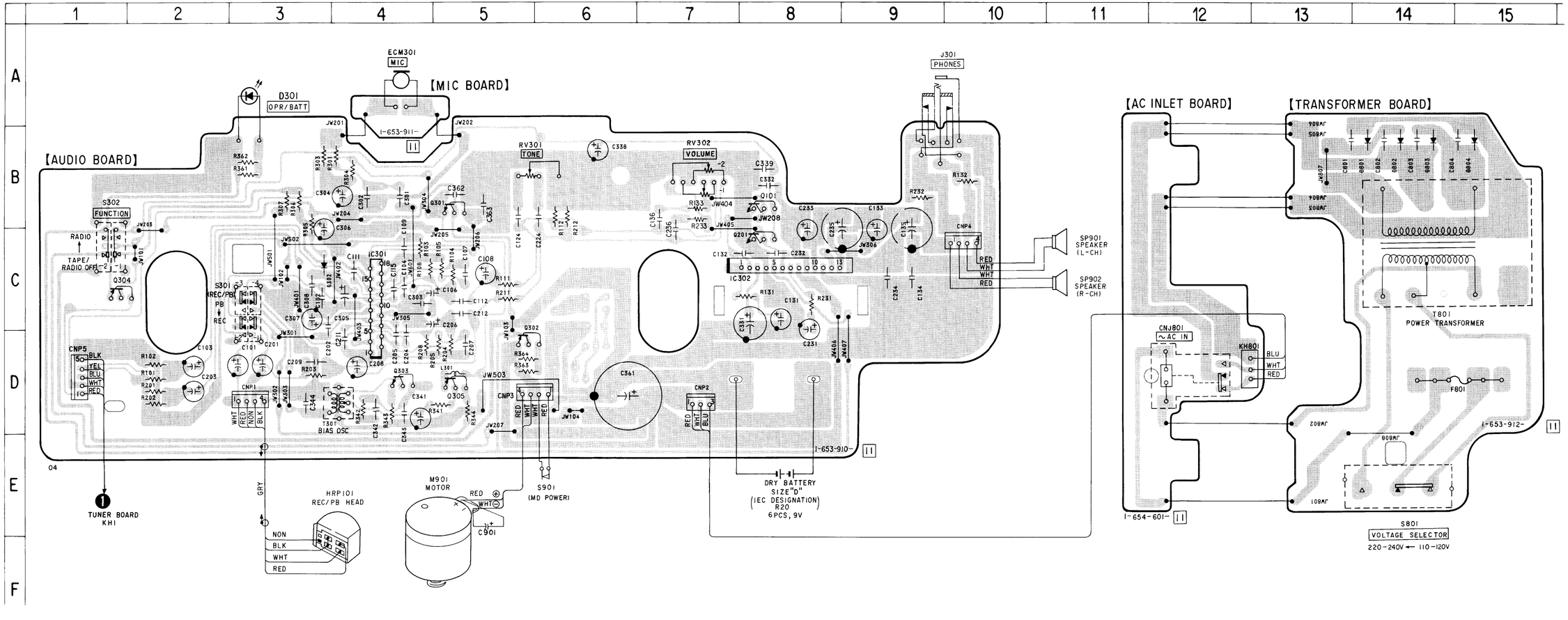
Note :

- All capacitors are in μF unless otherwise noted. pF : $\mu\mu$F 50WV or less are not indicated except for electrolytics and tantalums.
- All resistors are in Ω and 1/4 W or less unless otherwise specified.
- △ : internal component.
- B+ : B+ Line
- ▭ : adjustment for repair.
- Voltage is dc with respect to ground under no-signal (detuned) conditions.
- no mark: FM
  ( ): MW
  < >: SW
- Voltages are taken with a VOM (Input Impedance 10MΩ). Voltage variations may be noted due to normal production tolerances.
- Signal path.
  ⇨ : FM
- Abbreviation
  EA : Saudi Arabia model
  EE : East European model

4-3. PRINTED WIRING BOARDS—AUDIO SECTION—

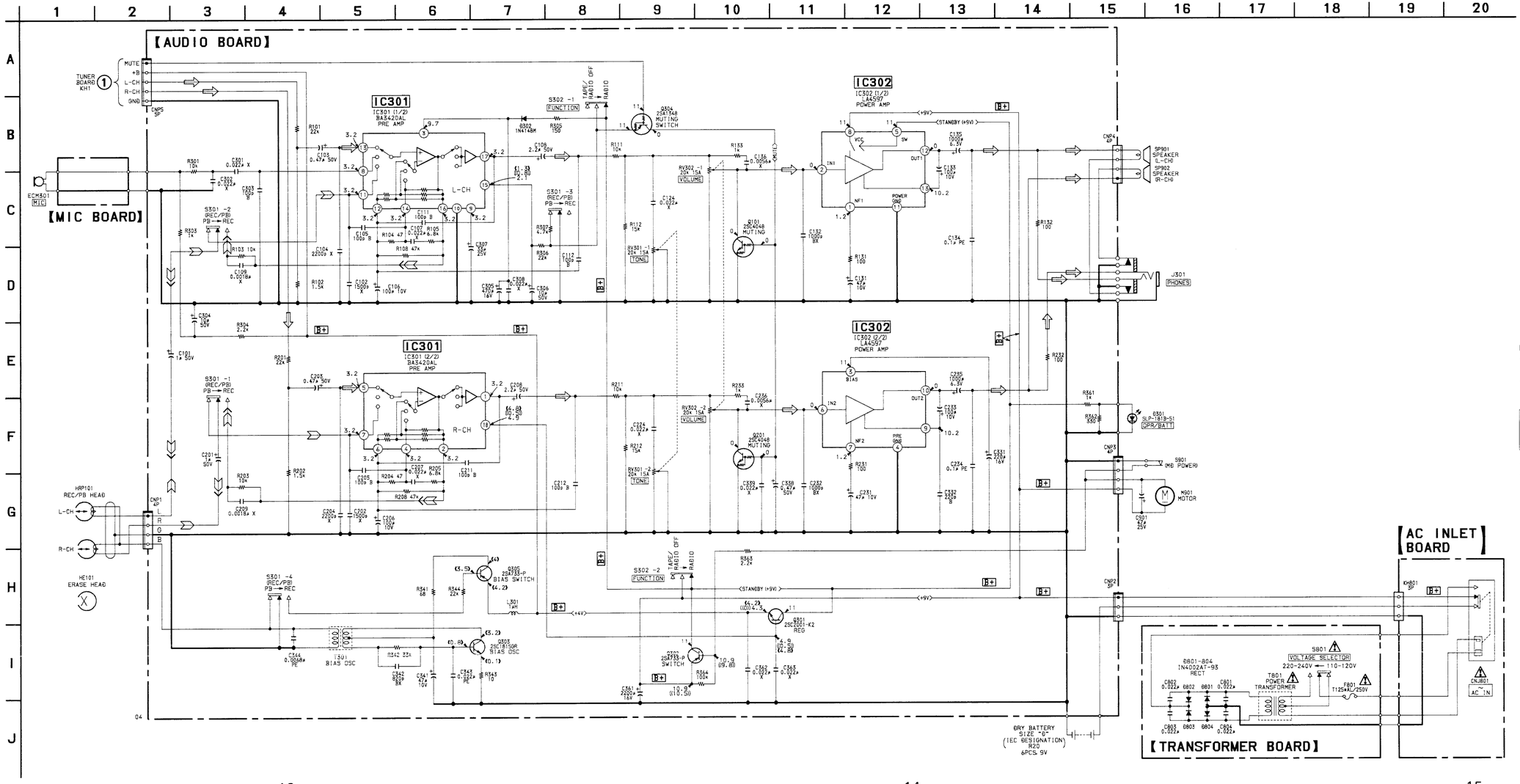
● Semiconductor Location

Ref. No.	Location
D301	A-3
D302	C-3
D801	B-14
D802	B-14
D803	B-14
D804	B-15
IC301	C-4
IC302	C-8
Q101	B-8
Q201	C-8
Q301	B-5
Q302	D-5
Q303	D-4
Q304	C-1
Q305	D-5



Note :
- ○— : parts extracted from the component side.
- ▨ : Pattern on the side which is seen.

4-4. SCHEMATIC DIAGRAM—AUDIO SECTION—



Note :

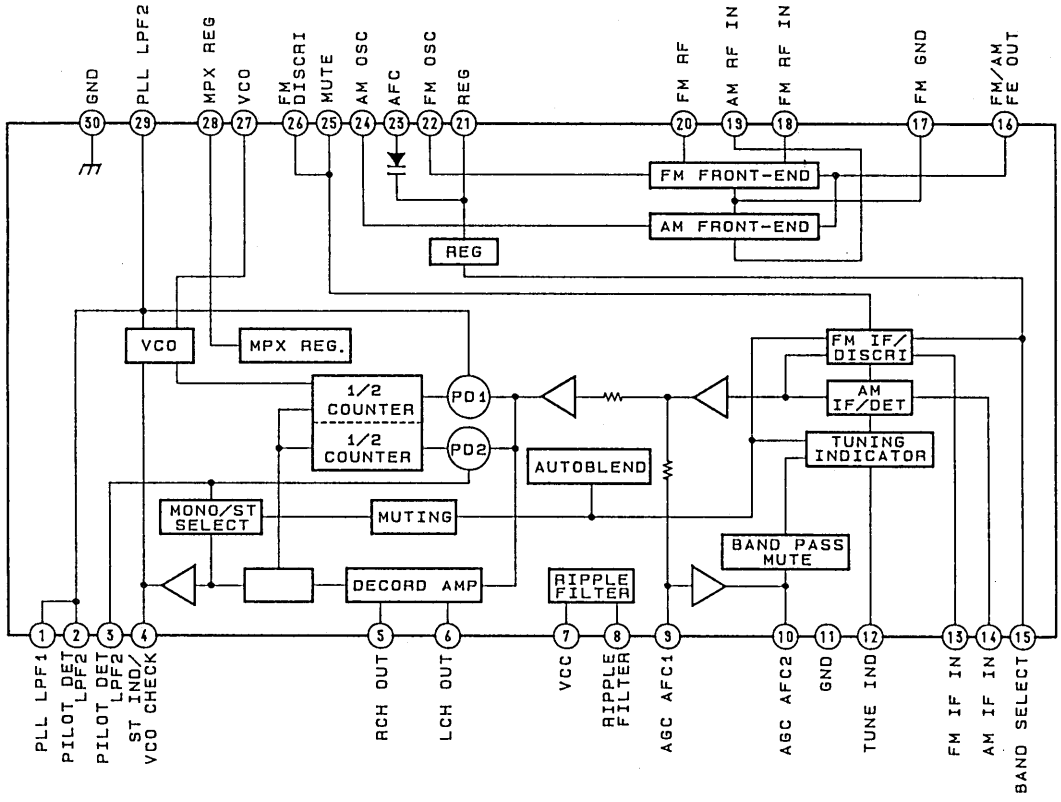
- All capacitors are in μF unless otherwise noted. pF : $\mu\mu$F 50WV or less are not indicated except for electrolytics and tantalums.
- All resistors are in Ω and 1/4 W or less unless otherwise specified.

Note : The components identified by mark ⚠ or dotted line with mark ⚠ are critical for safety. Replace only with part number specified.

- B+ : B+ Line
- Voltage is dc with respect to ground under no-signal (detuned) conditions.
- no mark: FM
  (( )): PB
  << >>: REC
- Voltages are taken with a VOM (Input Impedance 10MΩ). Voltage variations may be noted due to normal production tolerances.
- Signal path.
  ⇨ : FM
  ⇨ : PB
  ⇨ : REC

• **IC Block Diagram**

IC1 CXA1238S



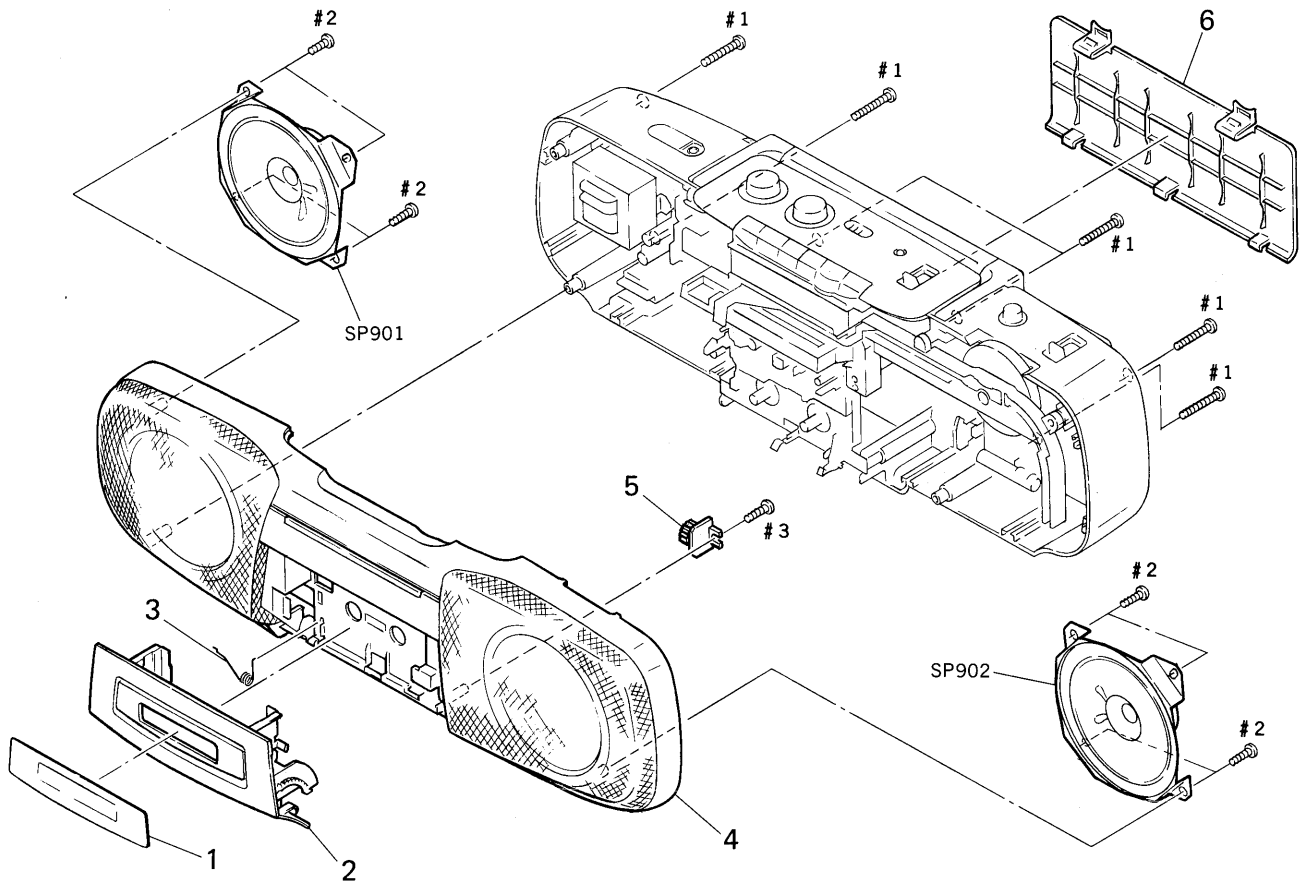
SECTION 5
EXPLODED VIEWS

NOTE:

- The mechanical parts with no reference number in the exploded views are not supplied.
- Items marked "*" are not stocked since they are seldom required for routine service. Some delay should be anticipated when ordering these items.
- Abbreviation
  EA : Saudi Arabia model
  UAE : United Arab Emirates model
  EE : East European model
- -XX and -X mean standardized parts, so they may have some difference from the original one.
- Color Indication of Appearance Parts
  Example :
  KNOB, BALANCE (WHITE)... (RED)
  ↑ ↑
  Parts Color Cabinet's Color
- Hardware (# mark) list and accessories and packing materials are given in the last of this parts list.

The components identified by mark ⚠ or dotted line with mark. ⚠ are critical for safety. Replace only with part number specified.

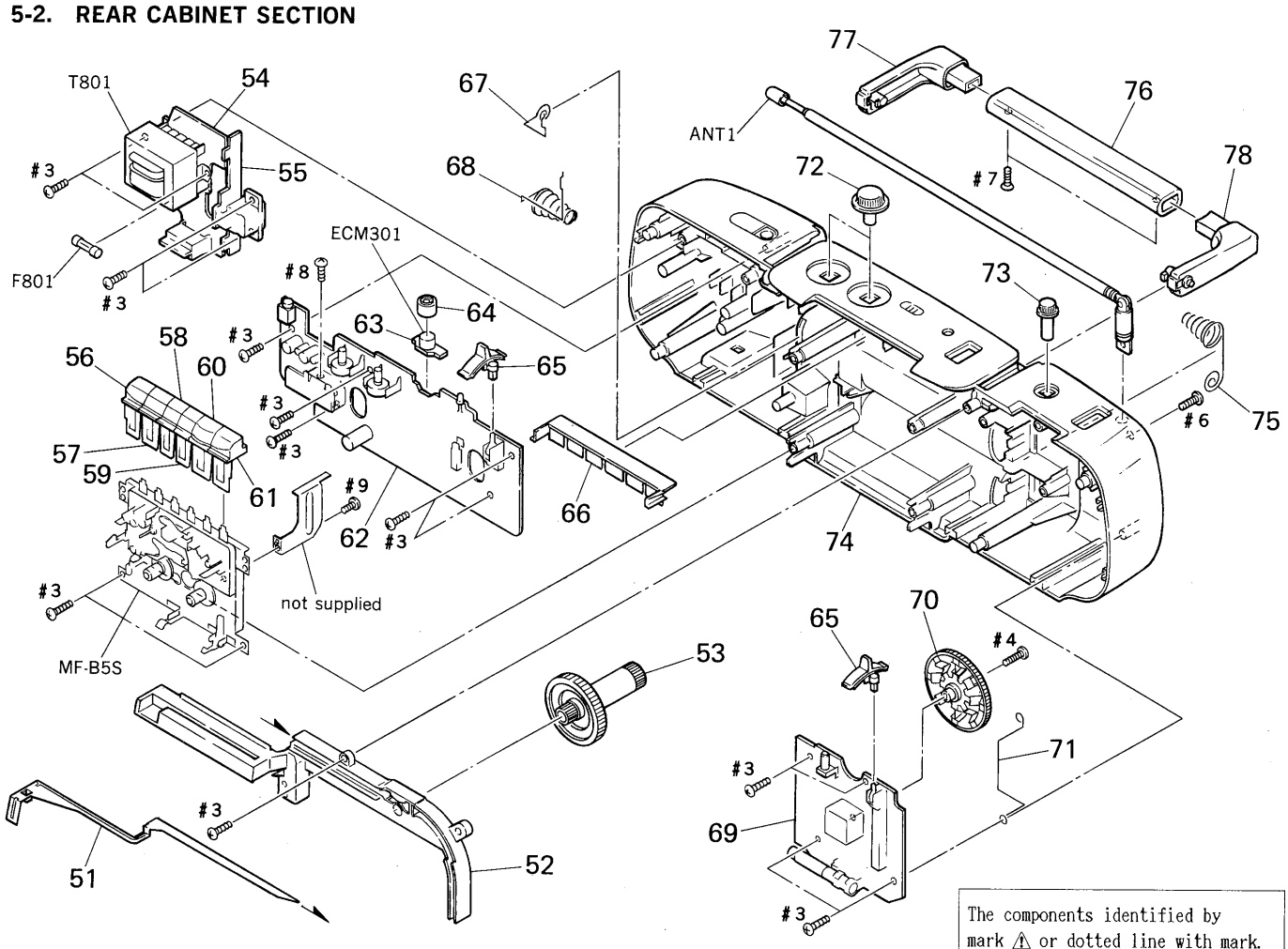
5-1. FRONT CABINET SECTION



Ref. No.	Part No.	Description	Remark
1	3-917-675-01	WINDOW, CASSETTE	
2	3-917-454-01	HOLDER (CASSETTE) (BLACK)	
2	3-917-454-11	HOLDER (CASSETTE) (RED) (EXCEPT EE)	
3	3-917-467-01	SPRING, CASSETTE	
4	X-3369-102-1	CABINET (FRONT) SUB ASSY (BLACK) (E, UAE)	
4	X-3369-103-1	CABINET (FRONT) SUB ASSY (BLACK) (EA)	
4	X-3369-426-1	CABINET (FRONT) SUB ASSY (RED) (E, UAE)	
4	X-3369-427-1	CABINET (FRONT) SUB ASSY (RED) (EA)	
4	X-3372-577-1	CABINET (FRONT) SUB ASSY (BLACK) (EE)	

Ref. No.	Part No.	Description	Remark
5	3-351-377-11	DAMPER	
6	3-917-646-01	LID, BATTERY CASE (BLACK)	
6	3-917-646-11	LID, BATTERY CASE (RED) (EXCEPT EE)	
SP901	1-504-684-11	SPEAKER (10CM) (L-CH)	
SP902	1-504-684-11	SPEAKER (10CM) (R-CH)	

5-2. REAR CABINET SECTION

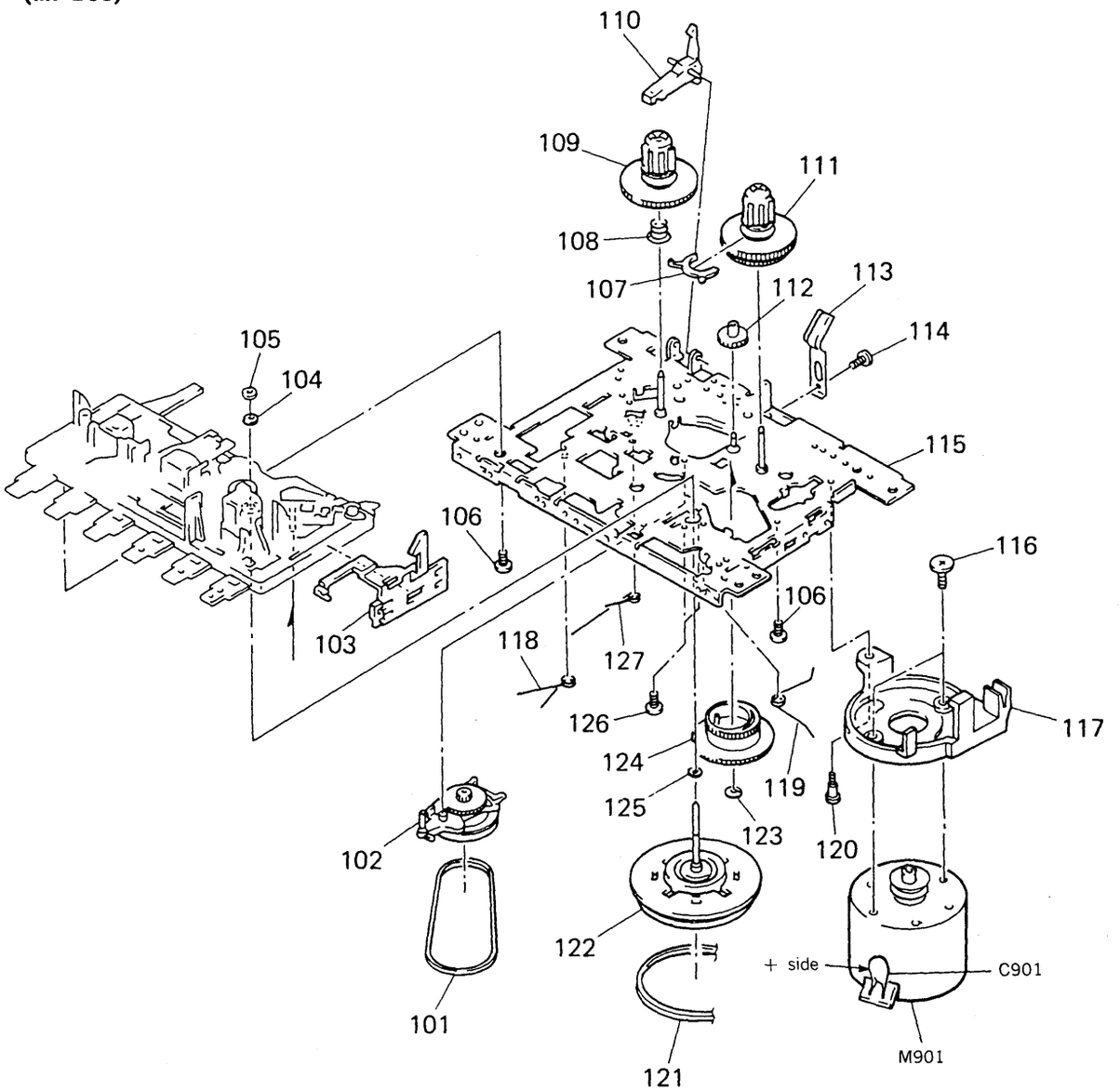


The components identified by mark ⚠ or dotted line with mark. ⚠ are critical for safety. Replace only with part number specified.

Ref. No.	Part No.	Description	Remark
51	3-917-463-01	POINTER	
* 52	3-917-461-01	CHASSIS (DIAL)	
53	3-917-465-01	KNOB (TUNING) (BLACK)	
53	3-917-465-11	KNOB (TUNING) (RED) (EXCEPT EE)	
* 54	1-653-912-11	TRANSFORMER BOARD	
* 55	1-654-601-11	AC INLET BOARD	
56	3-917-657-01	BUTTON (PAUSE)	
57	3-917-464-01	BUTTON (S/E)	
58	3-917-658-01	BUTTON (FF)	
59	3-917-659-01	BUTTON (REW)	
60	3-917-660-01	BUTTON (PLAY)	
61	3-917-661-01	BUTTON (REC)	
* 62	A-3269-575-A	AUDIO BOARD, COMPLETE	
* 63	1-653-911-11	MIC BOARD	
* 64	3-911-281-01	CUSHION, MIC	
65	3-917-650-01	KNOB (LEVER) (BLACK)	
65	3-917-650-11	KNOB (LEVER) (RED) (EXCEPT EE)	
* 66	3-917-649-01	BUTTON, CHASSIS MD (BLACK)	
* 66	3-917-649-11	BUTTON, CHASSIS MD (RED) (EXCEPT EE)	
67	3-917-654-01	TERMINAL (+), BATTERY	
68	3-917-655-01	TERMINAL (-), BATTERY	
* 69	A-3269-577-A	TUNER BOARD, COMPLETE (E, UAE)	
* 69	A-3269-578-A	TUNER BOARD, COMPLETE (EA)	
* 69	A-3306-013-A	TUNER BOARD, COMPLETE (EE)	

Ref. No.	Part No.	Description	Remark
70	3-917-466-01	DRUM (PVC)	
71	3-917-656-01	TERMINAL, ANTENNA	
72	3-917-647-01	KNOB (ROTARY) (BLACK)	
72	3-917-647-11	KNOB (ROTARY) (RED) (EXCEPT EE)	
73	3-917-648-01	KNOB (FINE TUNE) (BLACK)	
73	3-917-648-11	KNOB (FINE TUNE) (RED) (EXCEPT EE)	
* 74	3-917-452-01	CABINET (REAR) (BLACK) (E, EE)	
* 74	3-917-452-11	CABINET (REAR) (BLACK) (EA, UAE)	
* 74	3-917-452-21	CABINET (REAR) (RED) (E)	
* 74	3-917-452-31	CABINET (REAR) (RED) (EA, UAE)	
75	3-917-653-01	TERMINAL (+, -), BATTERY	
76	3-917-662-01	HANDLE (CENTER) (BLACK)	
76	3-917-662-11	HANDLE (CENTER) (RED) (EXCEPT EE)	
77	3-917-460-01	HANDLE (LEFT)	
78	3-917-459-01	HANDLE (RIGHT)	
ANT1	1-501-378-11	ANTENNA, TELESCOPIC	
ECM301	1-542-118-11	MICROPHONE, ELECTRET CONDENSER (MIC)	
⚠F801	1-532-611-00	FUSE, TIME-LAG (T125mAL/250V)	
⚠T801	1-450-015-11	TRANSFORMER, POWER	

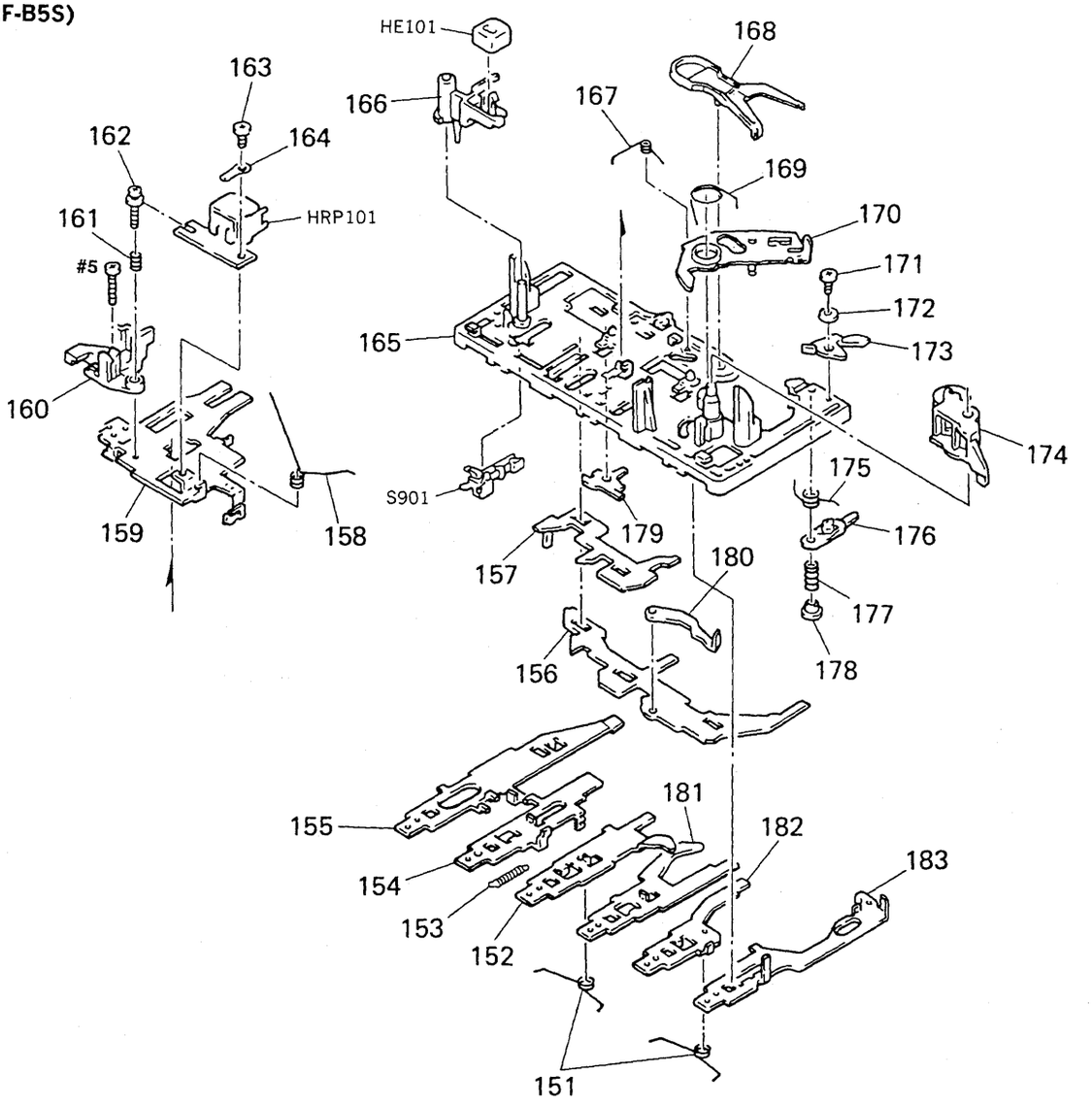
5-3. MECHANISM DECK SECTION (1) (MF-B5S)



Ref. No.	Part No.	Description	Remark
101	3-596-931-01	BELT, RF	
102	3-913-441-01	CLUTCH ASSY, RF	
103	3-348-366-01	LEVER, EJECT SLIDE	
104	3-913-442-01	WASHER, POLY-SLIDER	
105	3-913-434-01	WASHER, POLY-SLIDER	
106	3-348-370-01	SCREW (M2X5), P TIGHT BIND	
107	3-348-380-01	SENSOR	
108	3-913-429-01	SPRING, BACK TENSION	
109	3-913-430-01	REEL ASSY, SUPPLY	
110	3-348-368-01	LEVER, REC SAFETY	
111	3-913-431-01	REEL ASSY, TAKE-UP	
112	3-348-357-01	GEAR, FF	
113	3-348-367-01	PLATE, SP, PACK RETAINER	
114	3-913-444-01	SCREW (2X3)	
115	3-348-335-01	CHASSIS ASSY	

Ref. No.	Part No.	Description	Remark
116	3-598-855-01	MOTOR COLLER SCREW	
117	3-913-435-01	BRACKET, MOTOR	
118	3-348-337-01	SPRING	
119	3-348-336-01	SPRING	
120	3-395-995-01	SCREW, MB	
121	3-913-432-01	BELT, MAIN	
122	3-348-356-01	FLYWHEEL ASSY	
123	3-348-373-01	WASHER, POLYETHYLENE SLIDER	
124	3-348-353-01	GEAR, CAM	
125	3-913-433-01	WASHER, POLY-SLIDER	
126	3-348-371-01	SCREW (M2X4.5), CAMERA TAPPING	
127	3-348-341-01	SPRING	
C901	1-124-477-11	ELECT 47uF 20% 25V	
M901	X-3372-889-1	MOTOR ASSY	

5-4. MECHANISM DECK SECTION (2) (MF-B5S)



Ref. No.	Part No.	Description	Remark
151	3-348-334-01	SPRING (A)	
* 152	3-394-739-01	LEVER, REW BUTTON	
153	3-913-436-01	SPRING (S), PLAY BUTTON LEVER	
* 154	3-394-738-01	LEVER, PLAY BUTTON	
* 155	3-394-737-01	LEVER, REC BUTTON	
* 156	3-348-323-01	PLATE, FUNCTION, PUSH BUTTON	
* 157	3-348-322-01	PLATE, FUNCTION, SWITCH	
158	3-394-744-01	PANEL (P) (SP)	
* 159	3-395-986-01	PANEL, HEAD	
160	3-348-344-11	BASE, HEAD	
161	3-598-845-01	SPRING	
162	3-394-749-01	SCREW (M2X7), AZIMUTH	
163	3-348-372-01	SCREW (M2X3), + BIND	
164	3-394-748-01	LUG (B3)	
165	3-348-347-01	CHASSIS (MOLD) ASSY	
166	3-348-346-01	ARM, MG	
167	3-913-437-01	SPRING, M CONTROLLER	
168	3-913-439-01	LEVER, SENSING	

Ref. No.	Part No.	Description	Remark
169	3-348-351-01	SPRING	
* 170	3-913-440-01	PLATE ASSY, GEAR	
171	3-913-443-01	SCREW (2X3), PS TIGHT	
172	3-348-378-01	COLLAR, P ARM	
* 173	3-348-377-01	ARM, P	
174	3-913-438-01	ARM ASSY, PINCH ROLLER	
175	3-348-330-11	SPRING	
176	3-348-331-02	LEVER, PAUSE	
177	3-348-332-01	SPRING	
178	3-348-333-01	STOPPER, PAUSE	
179	3-348-340-01	STOPPER, PR	
* 180	3-348-339-01	LEVER, E KICK	
* 181	3-394-740-01	LEVER, FF BUTTON	
* 182	3-394-741-01	LEVER, STOP BUTTON	
* 183	3-394-742-01	LEVER, PAUSE BUTTON	
HE101	1-500-160-11	HEAD, ERASE	
HRP101	1-500-161-11	HEAD, RECORD/PLAYBACK	
S901	1-692-978-11	SWITCH, LEAF (MD POWER)	

SECTION 6
ELECTRICAL PARTS LIST

AC INLET **AUDIO**

NOTE:

- Due to standardization, replacements in the parts list may be different from the parts specified in the diagrams or the components used on the set.
- -XX and -X mean standardized parts, so they may have some difference from the original one.
- RESISTORS
  All resistors are in ohms.
  METAL:Metal-film resistor.
  METAL OXIDE: Metal oxide-film resistor.
  F:nonflammable
- Abbreviation
  EA : Saudi Arabia model
  UAE : United Arab Emirates model
  EE : East European model
  E23 : AC 110-120V area model
- Items marked "*" are not stocked since they are seldom required for routine service. Some delay should be anticipated when ordering these items.
- SEMICONDUCTORS
  In each case, u:μ, for example:
  uA..: μA.. uPA..: μPA..
  uPB..: μPB.. uPC..: μPC.. uPD..: μPD..
- CAPACITORS
  uF: μF
- COILS
  uH: μH

The components identified by mark ⚠ or dotted line with mark. ⚠ are critical for safety. Replace only with part number specified.

When indicating parts by reference number, please include the board.

Ref. No.	Part No.	Description			Remark
*	1-654-601-11	AC INLET BOARD			
		**********			
		< AC INLET >			
⚠CNJ801	1-526-838-11	INLET, AC 2P (~ AC IN)			
**********					
*	A-3269-575-A	AUDIO BOARD, COMPLETE			
		**********			
	7-685-871-01	SCREW +BVTT 3X6 (S)			
		< CAPACITOR >			
C101	1-124-903-11	ELECT	1uF	20%	50V
C102	1-162-301-11	CERAMIC	0.0015uF	20%	16V
C103	1-124-902-00	ELECT	0.47uF	20%	50V
C104	1-162-302-11	CERAMIC	0.0022uF	20%	16V
C105	1-162-282-31	CERAMIC	100PF	10%	50V
C106	1-124-443-00	ELECT	100uF	20%	10V
C107	1-162-843-11	CERAMIC	0.022uF	10%	16V
C108	1-126-961-11	ELECT	2.2uF	20%	50V
C109	1-162-830-11	CERAMIC	0.0018uF	10%	16V
C111	1-162-282-31	CERAMIC	100PF	10%	50V
C112	1-162-282-31	CERAMIC	100PF	10%	50V
C124	1-162-843-11	CERAMIC	0.022uF	10%	16V
C131	1-104-664-11	ELECT	47uF	20%	10V
C132	1-162-294-31	CERAMIC	0.001uF	10%	50V
C133	1-124-443-00	ELECT	100uF	20%	10V
C134	1-130-495-11	MYLAR	0.1uF	5%	50V
C135	1-124-471-00	ELECT	1000uF	20%	6.3V
C136	1-161-048-00	CERAMIC	0.0056uF	10%	16V
C201	1-124-903-11	ELECT	1uF	20%	50V
C202	1-162-301-11	CERAMIC	0.0015uF	20%	16V
C203	1-124-902-00	ELECT	0.47uF	20%	50V
C204	1-162-302-11	CERAMIC	0.0022uF	20%	16V
C205	1-162-282-31	CERAMIC	100PF	10%	50V
C206	1-124-443-00	ELECT	100uF	20%	10V
C207	1-162-843-11	CERAMIC	0.022uF	10%	16V
C208	1-126-961-11	ELECT	2.2uF	20%	50V
C209	1-162-830-11	CERAMIC	0.0018uF	10%	16V
C211	1-162-282-31	CERAMIC	100PF	10%	50V
C212	1-162-282-31	CERAMIC	100PF	10%	50V
C224	1-162-843-11	CERAMIC	0.022uF	10%	16V
C231	1-104-664-11	ELECT	47uF	20%	10V
C232	1-162-294-31	CERAMIC	0.001uF	10%	50V
C233	1-124-443-00	ELECT	100uF	20%	10V
C234	1-130-495-11	MYLAR	0.1uF	5%	50V
C235	1-124-471-00	ELECT	1000uF	20%	6.3V
C236	1-161-048-00	CERAMIC	0.0056uF	10%	16V
C301	1-162-843-11	CERAMIC	0.022uF	10%	16V
C302	1-162-843-11	CERAMIC	0.022uF	10%	16V
C303	1-162-282-31	CERAMIC	100PF	10%	50V
C304	1-124-907-11	ELECT	10uF	20%	50V
C305	1-126-941-11	ELECT	470uF	20%	16V
C306	1-124-907-11	ELECT	10uF	20%	50V
C307	1-126-966-11	ELECT	33uF	20%	25V
C308	1-162-843-11	CERAMIC	0.022uF	10%	16V
C331	1-124-120-11	ELECT	220uF	20%	25V
C332	1-162-286-21	CERAMIC	220PF	10%	50V
C338	1-124-902-00	ELECT	0.47uF	20%	50V
C339	1-162-843-11	CERAMIC	0.022uF	10%	16V
C341	1-104-664-11	ELECT	47uF	20%	10V
C342	1-162-293-31	CERAMIC	820PF	10%	50V
C343	1-130-487-00	MYLAR	0.022uF	5%	50V
C344	1-130-481-00	MYLAR	0.0068uF	5%	50V
C361	1-124-556-11	ELECT	2200uF	20%	16V
C362	1-162-843-11	CERAMIC	0.022uF	10%	16V
C363	1-162-843-11	CERAMIC	0.022uF	10%	16V
		< CONNECTOR >			
CNP1	1-506-986-11	PIN, CONNECTOR (PC BOARD) 4P			
* CNP2	1-580-155-11	PIN, CONNECTOR (PC BOARD) 3P			
CNP3	1-506-986-11	PIN, CONNECTOR (PC BOARD) 4P			
CNP4	1-506-986-11	PIN, CONNECTOR (PC BOARD) 4P			
CNP5	1-506-987-11	PIN, CONNECTOR (PC BOARD) 5P			

AUDIO | MIC | TRANSFORMER

Ref. No.	Part No.	Description			Remark
		< DIODE >			
D301	8-719-980-67	LED MVR3338S (OPR/BATT)			
D302	8-719-987-63	DIODE 1N4148M			
		< IC >			
IC301	8-759-978-54	IC BA3420AL			
IC302	8-759-820-22	IC LA4597			
		< JACK >			
J301	1-566-891-11	JACK (PHONES)			
		< COIL >			
L301	1-414-142-11	INDUCTOR	1uH		
		< TRANSISTOR >			
Q101	8-729-904-36	TRANSISTOR	DTC114YS		
Q201	8-729-904-36	TRANSISTOR	DTC114YS		
Q301	8-729-142-46	TRANSISTOR	2SC2001-LK		
Q302	8-729-119-76	TRANSISTOR	2SA1175-HFE		
Q303	8-729-281-53	TRANSISTOR	2SC1815-GR		
Q304	8-729-422-57	TRANSISTOR	UN4111		
Q305	8-729-119-76	TRANSISTOR	2SA1175-HFE		
		< RESISTOR >			
R101	1-247-863-00	CARBON	22K	5%	1/4W
R102	1-249-419-11	CARBON	1.5K	5%	1/4W
R103	1-249-429-11	CARBON	10K	5%	1/4W
R104	1-249-401-11	CARBON	47	5%	1/4W
R105	1-249-427-11	CARBON	6.8K	5%	1/4W
R108	1-249-437-11	CARBON	47K	5%	1/4W
R111	1-249-429-11	CARBON	10K	5%	1/4W
R112	1-249-431-11	CARBON	15K	5%	1/4W
R131	1-247-807-11	CARBON	100	5%	1/4W
R132	1-247-807-11	CARBON	100	5%	1/4W
R133	1-249-417-11	CARBON	1K	5%	1/4W
R201	1-247-863-00	CARBON	22K	5%	1/4W
R202	1-249-419-11	CARBON	1.5K	5%	1/4W
R203	1-249-429-11	CARBON	10K	5%	1/4W
R204	1-249-401-11	CARBON	47	5%	1/4W
R205	1-249-427-11	CARBON	6.8K	5%	1/4W
R208	1-249-437-11	CARBON	47K	5%	1/4W
R211	1-249-429-11	CARBON	10K	5%	1/4W
R212	1-249-431-11	CARBON	15K	5%	1/4W
R231	1-247-807-11	CARBON	100	5%	1/4W
R232	1-247-807-11	CARBON	100	5%	1/4W
R233	1-249-417-11	CARBON	1K	5%	1/4W
R301	1-249-429-11	CARBON	10K	5%	1/4W

Ref. No.	Part No.	Description			Remark
R303	1-249-417-11	CARBON	1K	5%	1/4W
R304	1-249-421-11	CARBON	2.2K	5%	1/4W
R305	1-249-407-11	CARBON	150	5%	1/4W
R306	1-247-863-00	CARBON	22K	5%	1/4W
R307	1-249-425-11	CARBON	4.7K	5%	1/4W
R341	1-249-403-11	CARBON	68	5%	1/4W
R342	1-249-435-11	CARBON	33K	5%	1/4W
R343	1-249-393-11	CARBON	10	5%	1/4W
R344	1-247-863-00	CARBON	22K	5%	1/4W
R361	1-249-417-11	CARBON	1K	5%	1/4W
R362	1-249-411-11	CARBON	330	5%	1/4W
R363	1-249-421-11	CARBON	2.2K	5%	1/4W
R364	1-249-441-11	CARBON	100K	5%	1/4W
		< VARIABLE RESISTOR >			
RV301	1-241-745-11	RES, VAR, CARBON 20K/20K (TONE)			
RV302	1-241-746-11	RES, VAR, CARBON 20K/20K (VOLUME)			
		< SWITCH >			
S301	1-571-472-11	SWITCH, SLIDE (REC/PB)			
S302	1-570-729-11	SWITCH, LEVER SLIDE (FUNCTION)			
		< TRANSFORMER >			
T301	1-433-268-00	TRANSFORMER, BIAS OSCILLATOR			
**********					
*	1-653-911-11	MIC BOARD			
		*********			
		< MICROPHONE >			
ECM301	1-542-118-11	MICROPHONE, ELECTRET CONDENSER (MIC)			
**********					
*	1-653-912-11	TRANSFORMER BOARD			
		**********			
	1-533-217-31	HOLDER, FUSE			
		< CAPACITOR >			
C801	1-101-005-00	CERAMIC	22000PF		50V
C802	1-101-005-00	CERAMIC	22000PF		50V
C803	1-101-005-00	CERAMIC	22000PF		50V
C804	1-101-005-00	CERAMIC	22000PF		50V
		< DIODE >			
D801	8-719-200-02	DIODE 10E2			
D802	8-719-200-02	DIODE 10E2			
D803	8-719-200-02	DIODE 10E2			
D804	8-719-200-02	DIODE 10E2			

Ref. No.	Part No.	Description			Remark
		< FUSE >			
⚠F801	1-532-611-00	FUSE, TIME-LAG (T125mAL/250V)			
		< SWITCH >			
⚠S801	1-552-921-00	SWITCH, POWER (VOLTAGE SELECTOR)			
		< TRANSFORMER >			
⚠T801	1-450-015-11	TRANSFORMER, POWER			
**********					
*	A-3269-577-A	TUNER BOARD, COMPLETE (E, UAE)			
*	A-3269-578-A	TUNER BOARD, COMPLETE (EA)			
*	A-3306-013-A	TUNER BOARD, COMPLETE (EE)			
		**********			
	3-917-466-01	DRUM (PVC)			
	7-621-770-87	SCREW +P 2.6X5			
		< CAPACITOR >			
C1	1-162-286-21	CERAMIC	220PF	10%	50V
C2	1-162-286-21	CERAMIC	220PF	10%	50V
C3	1-162-306-11	CERAMIC	0.01uF	20%	16V
C4	1-124-907-11	ELECT	10uF	20%	50V
C5	1-162-306-11	CERAMIC	0.01uF	20%	16V
C6	1-162-198-31	CERAMIC	8.2PF	10%	50V (EE)
C6	1-162-205-31	CERAMIC	18PF	5%	50V (E, UAE)
C6	1-162-209-31	CERAMIC	27PF	5%	50V (EA)
C7	1-162-187-31	CERAMIC	1PF	20%	50V (EXCEPT EA)
C8	1-102-947-00	CERAMIC	10PF	5%	50V (EE)
C8	1-102-953-00	CERAMIC	18PF	5%	50V (E, UAE)
C8	1-102-959-00	CERAMIC	22PF	5%	50V (EA)
C10	1-162-193-31	CERAMIC	3.3PF	10%	50V
C11	1-162-306-11	CERAMIC	0.01uF	20%	16V
C13	1-162-196-31	CERAMIC	5.6PF	10%	50V (EXCEPT EA)
C13	1-162-197-31	CERAMIC	6.8PF	10%	50V (EA)
C14	1-162-282-31	CERAMIC	100PF	10%	50V
C15	1-126-963-11	ELECT	4.7uF	20%	50V
C16	1-124-903-11	ELECT	1uF	20%	50V
C17	1-104-664-11	ELECT	47uF	20%	10V
C18	1-104-664-11	ELECT	47uF	20%	10V
C19	1-162-306-11	CERAMIC	0.01uF	20%	16V
C20	1-124-903-11	ELECT	1uF	20%	50V
C21	1-126-961-11	ELECT	2.2uF	20%	50V
C22	1-124-443-00	ELECT	100uF	20%	10V
C23	1-124-902-00	ELECT	0.47uF	20%	50V
C24	1-126-963-11	ELECT	4.7uF	20%	50V
C25	1-126-956-11	ELECT	0.1uF	20%	50V
C27	1-162-841-11	CERAMIC	0.015uF	10%	16V
C28	1-162-841-11	CERAMIC	0.015uF	10%	16V
C32	1-162-197-31	CERAMIC	6.8PF	10%	50V
C34	1-162-306-11	CERAMIC	0.01uF	20%	16V
C35	1-162-306-11	CERAMIC	0.01uF	20%	16V
C36	1-164-616-11	CERAMIC	4PF	5%	50V
C38	1-164-617-11	CERAMIC	4PF	5%	50V
C39	1-102-949-00	CERAMIC	12PF	5%	50V
C40	1-136-480-11	FILM	0.0015uF	5%	100V
C41	1-136-683-11	FILM	0.0056uF	5%	100V
C42	1-107-734-11	FILM	360PF	5%	100V
C43	1-162-306-11	CERAMIC	0.01uF	20%	16V
C45	1-162-282-31	CERAMIC	100PF	10%	50V
C46	1-162-282-31	CERAMIC	100PF	10%	50V
C47	1-162-282-31	CERAMIC	100PF	10%	50V
C50	1-162-282-31	CERAMIC	100PF	10%	50V
C51	1-162-294-31	CERAMIC	0.001uF	10%	50V
C52	1-162-202-31	CERAMIC	13PF	5%	50V
C53	1-162-287-31	CERAMIC	270PF	10%	50V
C71	1-162-306-11	CERAMIC	0.01uF	20%	16V
		< FILTER >			
CF1	1-577-327-81	FILTER, CERAMIC			
CF2	1-567-670-11	FILTER, CERAMIC			
CF3	1-577-327-81	FILTER, CERAMIC			
CF4	1-577-327-81	FILTER, CERAMIC			
		< VARIABLE CAPACITOR >			
CT1-4	1-151-693-11	CAP, VAR (TUNING) (EXCEPT EE)			
CV1-4	1-151-693-11	CAP, VAR (TUNING) (EXCEPT EE)			
CT1-4	1-151-710-11	CAP, VAR (TUNING) (EE)			
CV1-4	1-151-710-11	CAP, VAR (TUNING) (EE)			
		< TRIMMER >			
CT6	1-141-485-11	CAP, ADJ 5PF			
CT10	1-141-485-11	CAP, ADJ 5PF			
		< DIODE >			
D1	8-719-987-63	DIODE 1N4148M			
D2	8-719-987-63	DIODE 1N4148M			
D3	8-719-987-63	DIODE 1N4148M			

The components identified by mark ⚠ or dotted line with mark. ⚠ are critical for safety. Replace only with part number specified.

TUNER

Ref. No.	Part No.	Description	Remark
		< FILTER >	
FL1	1-239-333-11	FILTER, BAND PASS (EXCEPT EE)	
FL1	1-239-831-11	FILTER, BAND PASS (EE)	
		< IC >	
IC1	8-752-050-20	IC CXA1238S	
		< COIL >	
L1	1-406-998-11	COIL, AIR-CORE (E, UAE)	
L1	1-422-317-31	COIL, AIR-CORE (EA)	
* L1	1-428-212-11	COIL, AIR-CORE (EE)	
L2	1-402-683-11	COIL (OSC) (EE)	
L2	1-406-868-11	COIL, AIR-CORE (E, UAE)	
L2	1-406-956-11	COIL (WITH CORE) (EA)	
L3	1-402-593-11	ANTENNA, FERRITE-ROD (MW)	
L4	1-406-253-11	COIL (OSC)	
L5	1-402-597-31	COIL (ANT)	
L6	1-406-247-11	COIL (OSC)	
L7	1-402-595-31	COIL (ANT)	
L8	1-406-246-11	COIL (OSC)	
L9	1-410-314-11	INDUCTOR 0.47uH	
L11	1-410-315-21	INDUCTOR 0.68uH	
L12	1-410-314-11	INDUCTOR 0.47uH	
		< TRANSISTOR >	
Q2	8-729-281-53	TRANSISTOR 2SC1815-GR	
		< RESISTOR >	
R3	1-249-421-11	CARBON 2.2K 5% 1/4W	
R6	1-249-441-11	CARBON 100K 5% 1/4W	
R9	1-249-441-11	CARBON 100K 5% 1/4W	
R11	1-249-427-11	CARBON 6.8K 5% 1/4W	
R12	1-249-427-11	CARBON 6.8K 5% 1/4W	
R13	1-247-807-11	CARBON 100 5% 1/4W	
R14	1-249-429-11	CARBON 10K 5% 1/4W	
R15	1-247-887-00	CARBON 220K 5% 1/4W	
R16	1-247-887-00	CARBON 220K 5% 1/4W	
R17	1-249-421-11	CARBON 2.2K 5% 1/4W	
R18	1-249-417-11	CARBON 1K 5% 1/4W	
R22	1-249-411-11	CARBON 330 5% 1/4W	
R30	1-249-417-11	CARBON 1K 5% 1/4W	
R31	1-247-854-11	CARBON 9.1K 5% 1/4W	
R32	1-249-441-11	CARBON 100K 5% 1/4W	
		< VARIABLE RESISTOR >	
RV1	1-223-641-11	RES, ADJ, CARBON 22K	
RV2	1-223-384-11	RES, VAR, CARBON 100K (FINE TUNE)	

Ref. No.	Part No.	Description	Remark
		< SWITCH >	
S1	1-571-147-11	SWITCH, LEVER SLIDE (BAND)	
		< TRANSFORMER >	
T1	1-406-952-11	TRANSFORMER, IF	
**********			
		MISCELLANEOUS	
		**********	
ANT1	1-501-378-11	ANTENNA, TELESCOPIC	
C901	1-124-477-11	ELECT 47uF 20% 25V	
⚠F801	1-532-611-00	FUSE, TIME-LAG (T125mAL/250V)	
HE101	1-500-160-11	HEAD, ERASE	
HRP101	1-500-161-11	HEAD, RECORD/PLAYBACK	
M901	X-3372-889-1	MOTOR ASSY	
S901	1-692-978-11	SWITCH, LEAF (MD POWER)	
SP901	1-504-684-11	SPEAKER (10CM) (L-CH)	
SP902	1-504-684-11	SPEAKER (10CM) (R-CH)	
⚠T801	1-450-015-11	TRANSFORMER, POWER	
**********			
		ACCESSORIES & PACKING MATERIALS	
		**********	
⚠	1-551-983-00	CORD, POWER (E23)	
⚠	1-569-007-11	ADAPTOR, CONVERSION 2P (E23)	
⚠	1-569-008-11	ADAPTOR, CONVERSION 2P (EA)	
⚠	1-590-413-11	CORD, POWER (EXCEPT E23, EE)	
⚠	1-769-412-11	CORD, POWER (EE)	
	3-759-321-11	MANUAL, INSTRUCTION (ENGLISH, FRENCH, SPANISH) (EXCEPT EE)	
	3-759-321-21	MANUAL, INSTRUCTION (POLISH, ENGLISH, GERMAN, RUSSIAN) (EE)	
*	3-917-148-01	INDIVIDUAL CARTON (EXCEPT EE)	
*	3-917-494-01	CUSHION (L)	
*	3-917-495-01	CUSHION (R)	
*	3-939-150-01	INDIVIDUAL CARTON (EE)	
**********			

The components identified by mark ⚠ or dotted line with mark. ⚠ are critical for safety. Replace only with part number specified.

Ref. No.	Part No.	Description	Remark

****
HARDWARE LIST
****

#1	7-685-651-79	SCREW +BVTP 3X20 TYPE2	
#2	7-685-646-79	SCREW +BVTP 3X8 TYPE2 N-S	
#3	7-685-647-79	SCREW +BVTP 3X10 TYPE2 N-S	
#4	7-621-770-87	SCREW +P 2.6X5	
#5	7-621-255-45	SCREW +P 2X6	
#6	7-682-548-09	SCREW +B 3X8	
#7	7-685-248-19	SCREW +KTP 3X12 TYPE2 SLIT	
#8	7-685-871-01	SCREW +BVTT 3X6 (S)	
#9	7-685-101-11	SCREW +PTP 2X3 NON-SLIT	

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SONY®
SERVICE MANUAL

E Model

SUPPLEMENT-1

File this supplement with the service manual.

Subject : Add the Hong Kong Model

The Hong Kong model is the same as the E model except for the accessories & packing materials.
See the information on E model for details.

Part No.	Description	Remark
⚠1-770-019-11	ADAPTOR, CONVERSION	
⚠1-696-820-21	CORD, POWER	
*3-923-673-01	INDIVIDUAL CARTON	

NOTE :

- Items marked "*" are not stocked since they are seldom required for routine service. Some delay should be anticipated when ordering these items.

The components identified by mark ⚠ or dotted line with mark. ⚠ are critical for safety. Replace only with part number specified.

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Consumer A & V Products Company
Personal A & V Products Div.

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